

24BCON0871

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Assignment-3

Section - A

1. Define routing in computer networks.

Ans) Routing is the process of selecting the best path for data packets to travel across a network, from the source to destination. It is performed by routers using routing tables & protocols to determine the most efficient path based on metrics like hop count, cost or delay.

2. Differentiate b/w static routing and dynamic routing.

Ans)

Basis	Static Routing	Dynamic Routing
Definition	Routes are manually configured by the network administrator.	Routes are automatically updated using routing protocols.
Update	No automatic updates	Updates automatically as network topology changes.
Complexity	Simple & suitable for small networks	Complex but ideal for large, changing networks.
Example	Configured using "route" command.	OSPF, RIP and EIGRP.

3. Write a short note on ARP and RARP.

Ans) ARP (Address Resolution Protocol):

It maps an IP address to its corresponding MAC address so that data can be sent on a local network.

RARP (Reverse Address Resolution Protocol):

It is used by a device to request its IP address.

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from a gateway when only the MAC address is known.

4. Explain the role of DHCP in computer network.

Ans) DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, gateway, & DNS information to device in a network. This reduces manual configuration and avoids IP conflicts, making network management efficient.

5. What is Network Address Translation (NAT) & why is it used?

Ans) NAT converts private IP addresses used within a local network to public IP address for communication over the internet.

Uses:-

- Conserves public IP addresses.
- Provides security by hiding internal IPs.
- Enables multiple devices to share one public IP.

Section - B

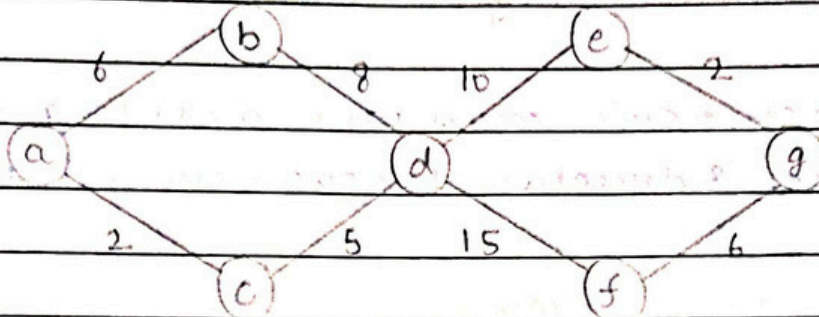
1. Explain shortest path routing algorithm with a suitable example.

Ans) It refers to the algorithm that helps to find the shortest path between a sender and receiver for routing the data packets through the network in terms of shortest distance, minimum cost, and minimum time.

→ Common Shortest Path Algorithms:-

- Dijkstra's Algorithm.
- Bellman Ford's Algorithm
- Floyd Warshall's Algorithm

Example:-



- (i) Start from node a & mark node a as visited and check for adjacent nodes.

Distance: a b c d e f g

0	∞	∞	∞	∞	∞	∞
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Unvisited Nodes: ~~a, b, c, d, e, f, g~~ {a, b, c, d, e, f, g}

- (ii) Now check adjacent node with light weight and move to that node.

Distance: a b c d e f g

0	∞	∞	∞	∞	∞	∞
0	6	2	∞	∞	∞	∞

- (iii) Mark node ^c as visited and move to next node d.

Distance: a b c d e f g

0	∞	∞	∞	∞	∞	∞
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0	6	2	∞	∞	∞	∞
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0	6	2	7	∞	∞	∞
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- (iv) Similarly we will normalize other node considering it should not form a cycle and will keep track is visited node.

(v) Mark node g as visited and add the distance.

Distance: a b c d e f g
 0 2 6 7 17 22 19

2. Differentiate b/w Distance vector and link state routing protocols.

Ans) → Distance Vector Routing Protocol

- Concept

In this, each router maintains a routing table containing the distance & direction to reach every destination in network.

- Working

Each router periodically shares its entire routing table with direct neighbours. When routers receive updates from their neighbours, they calculate shortest path to all destinations using algorithm like Bellman Ford.

- Update Mechanism

Updates occur periodically.

- Example:

↳ RIP (Routing Information Protocol)

↳ IGRP (Interior Gateway Routing Protocol)

→ Link State Routing Protocol

- Concept

Each router builds a complete map of network & independently computes the shortest path to every other router using Dijkstra's Shortest Path Algorithm.

- Working

Each router discovers its neighbour & measures the cost

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to reach them.

- Update Mechanism

Updates are sent only when a change in the topology occurs, not periodically.

- Example:

↳ OSPF (Open Shortest Path First)

↳ IS-IS (Intermediate System to Intermediate System).

3. Write a short note on ICMP & its functions in the internet.

Ans) ICMP (Internet Control Message Protocol) is a network layer protocol used by network devices like routers & hosts to send control messages, error reports, status information.

→ Purpose

ICMP is not used for data transmission, but it helps in:

- Reporting network issues.
- Testing connectivity between devices.
- Informing about delivery or routing failures.

→ Working

- ICMP messages are encapsulated directly within IP datagrams.
- When a network device encounters a problem in processing a packet, it sends an ICMP message back to source IP address.
- These messages help network administrators diagnose & troubleshoot network problems.

→ Example:

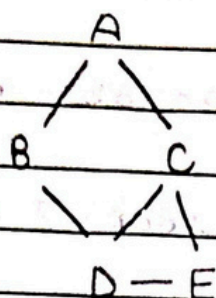
- When you use the ping command:
- ↳ The source host sends an ICMP Echo Request message to destination.
- ↳ The destination replies with ICMP Echo Reply message.
- ↳ The round-trip time is calculated to test the connection.

Section - C

1. Explain in detail various routing algorithms such as flooding, OSPF and Path Vector with suitable diagrams.

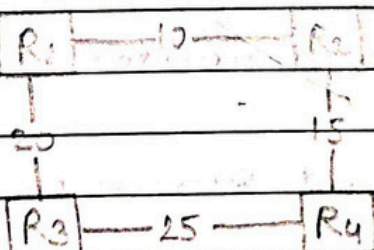
Ans) i) Flooding Algorithm

- Every incoming packet is sent out of an every outgoing line except the one on which it arrived.
- Working :-
 - ↳ Each node receives a packet.
 - ↳ It forwards the packet to all its neighbours except the one from which it received it.
 - ↳ To avoid infinite looping, each packet has to hop count or sequence number.
- Advantages :-
 - ↳ Simple and reliable.
 - ↳ Ensure that packet reach all destinations.
- Disadvantages :-
 - ↳ Generates redundant traffic → consumes bandwidth.
- Diagram:



(ii) OSPF (Open Shortest Path First)

- Link-state Routing Protocol (used in IP networks).
- Working :-
 - ↳ Each router discovers its neighbours.
 - ↳ It measures the cost to each neighbour.
 - ↳ Build a link state database (LSDB) shared with all routers.
 - ↳ Each routers independently computes ~~short~~ shortest path tree using Dijkstra's Algorithm.
- Advantages :-
 - ↳ Fast Convergence.
 - ↳ Scalable for large networks.
- Disadvantages :-
 - ↳ Complex Implementation
- Diagram:



(iii) Path Vector Algorithm

- Each router maintains the entire path that data must traverse to reach a destination.

- Working :-

- ↳ Each router advertise its ad reachable networks & complete path.
- ↳ Routers use these paths to avoid loops & select optimal routes.

- Advantages :-

- ↳ Prevents routing loops.
- ↳ Scalable for inter-domain routing.

- Disadvantages :-

- ↳ Path information can be large.

- Diagram :-

AS1 --- AS2 --- AS3 --- AS4

Path from AS1 to AS4 : [AS1, AS2, AS3, AS4]

2. What is hierarchical routing? Explain broadcast and multicast routing with diagrams.

Ans) Hierarchical Routing

- A routing method that divides the network into regions or domains, reducing the size of routing table & improving scalability.

- Working :-

- ↳ Each router knows routes within its region in detail.
- ↳ For other regions, it knows only path to reach the region.

- Advantages :-

- ↳ Reduce memory & computation overhead.
- ↳ Simplifies routing in large networks.

Region 1

Region 2

R₁ — R₂R₃ — R₄

Gateway

Broadcast Routing

- Sending a packet from one source to all nodes in network.
- Techniques:
 - ↳ Flooding - Each node sends to all neighbors.
 - ↳ Spanning Tree - Builds a loop free path to all nodes.
- Use Case - ARP request (Address Resolution Protocol)

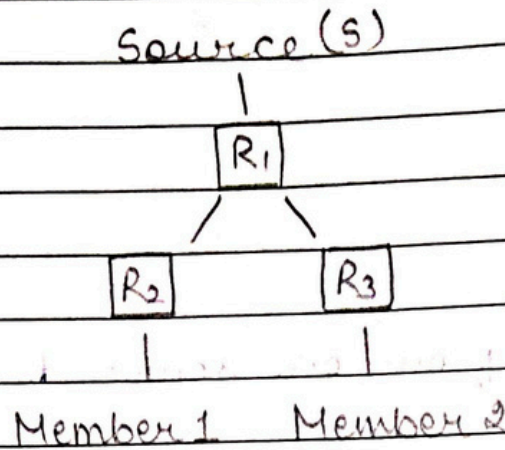
All nodes receive same message.

Multicast Routing

- Sending data from one source to a selected group of destination.
- Working:-
 - ↳ Routers create multicast tree that connects all members of group.
 - ↳ Uses protocol like IGMP and PIM.
- Advantages:-

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- ↳ efficient bandwidth usage.
- ↳ Used in video conferencing, online streaming.



3. Describe IP protocol in detail. Explain the structure of an IP address. Differentiate b/w class A, B, C, D, E with suitable examples.

Ans:

The internet protocol is a core network protocol responsible for addressing and routing ^{etc} packages of data between computers across interconnected networks. IP ensures data moves efficiently from source to destination by assigning unique identifiers & managing packet delivery through routine mechanisms.

→ IP Address Structure.

An IP address is a numerical label for each device on a network, used to uniquely identify that device & allow communication. In IPv4, an IP address is 32 bit long, typically represented as four decimal number (octets) separated by dots, for example: 192.168.1.15. Each octet range from 0 to 255.

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- Binary form - Split into four 8-bit segments (octets).
- Network ID - Specifies the unique network.
- Host ID - Identifies the specific device within that network.

Example: For 192.168.1.15

Network ID - 192.168.1

Host ID - 15

→ IP Address Classes

IPv4 addresses are divided into five classes (A, B, C, D, E), distinguished by the value of first octet & ~~serve~~ serving different purposes.

Class	First Octet Range	Network/Host Division	Example
A	1-126	8 bit Network/24 bit Host	10.50.120.7
B	128-191	16 bit Network/16 bit Host	172.16.55.13
C	192-223	24 bit Network/8 bit Host	192.168.5.1
D	224-239	Reserved for Multicast	224.10.10.10
E	240-255	Experimental	250.1.1.1